

Vision-Based Continuous Sign Language Spotting Using Gaussian Hidden Markov Model

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Abstract—The presence of movement epenthesis (*me*) between two consecutive signs in a continuous sign language makes sign language recognition (SLR) a challenging task. In this letter, we propose a vision-based continuous SL spotting system, which separates the meaningful signs from the sign sequences by removing the *me* components from H.264/AVC compressed videos. The work is based on a two-state hidden Markov model (HMM) with Gaussian emission probability. The HMM is trained by the feature set extracted from the entire sign sequence video and finally, the hidden state-sequence is decoded using the Viterbi algorithm. From the decoded state-sequence, the sign spotting is done. The feature set comprises features extracted from the compressed domain as well as uncompressed domain analysis of the sign video. The video database is composed of American SL videos collected from Boston University database. From the experimental results, it is seen that the proposed system can spot the sign and *me* frames with a spotting rate of about 83%.

Index Terms—Sensor applications, continuous sign language, hidden Markov model (HMM), movement epenthesis, sign spotting.

I. INTRODUCTION

Sign language is a visual language used by the deaf community. The aim of the sign language recognition (SLR) system is to transcribe sign language into text or speech so that it can be easily understood by the hearing community [1]. In continuous SL, there can be many nonsign movements (called *movement epenthesis*) between true signs which connect the end location of the previous sign to the start location of the next sign [2]. Fig. 1 shows the movement epenthesis (*me*) frames between the end location of sign “GATE” and the start location of the sign “WHERE” in a continuous sign video [2]. So, the sign spotting, i.e., the process of segmenting out the signs from continuous utterance sentences by removing the *me* components is a challenging task for continuous SLR systems [3].

The difficulty in sign language spotting is that instances of signs vary in both motion and appearance [4]. Hidden Markov model (HMM) is useful for dealing with the time-varying nature of features [5], [6]. Mohandes *et al.* [6] proposed a signer-independent Arabic SLR system where HMM was used to present the time-varying feature patterns. Azar and Seyedarabi [5] presented work on a dynamic Persian SLR system using HMM. Kumar *et al.* [7] proposed a multisensor fusion framework for SLR using coupled HMM (CHMM) for recognition of dynamic isolated sign gestures. In [1], Gao *et al.* proposed a signer-independent continuous SLR system using the self-organizing feature maps (SOFM) as different signer’s feature extractor for continuous HMM. Yang *et al.* [8] proposed a continuous SLR using level building based on HMM. Some works were also reported using dynamic time warping (DTW) [9], [10] which is a widely used algorithm for template-matching based approaches. Recently, deep learning (DL)-based frameworks have been implemented for continuous SLR [11], [12]. But both approaches are computationally expensive and require huge databases.

Generally, for SLR, the first and third problems of HMM, i.e., evaluation and fitting [13] are given more emphasis. But in our work, the second problem of the HMM, i.e., inference is extensively used. The contributions of the letter are as follows:

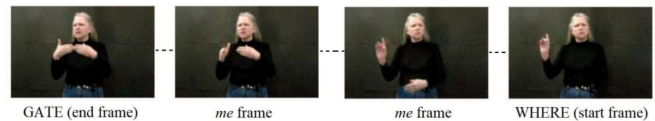


Fig. 1. Movement epenthesis (*me*) between “GATE” and “WHERE” signs [2].

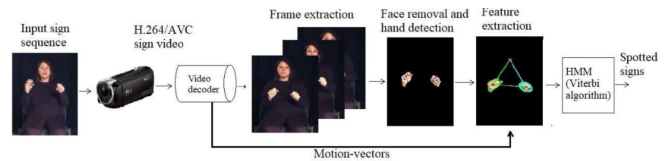


Fig. 2. Proposed HMM-based framework for continuous sign language spotting.

- 1) We propose a continuous sign language spotting framework using a two-state HMM in a novel way.
- 2) A fusion of both compressed as well as spatial domain features is used.

The rest of this letter is organized as follows. Section II describes the proposed work, Section III describes the experimental results obtained, and finally, Section IV concludes this letter.

II. PROPOSED GAUSSIAN HMM FOR CONTINUOUS SIGN LANGUAGE SPOTTINGS

The proposed framework for continuous SL spotting using Gaussian HMM is shown in Fig. 2. First, the input sign sequence is captured with an RGB camera. Then the captured H.264/AVC video is decoded to extract the frames along with the motion vectors (MVs) that are inherent in the compressed video itself. After that, the face of the signer is detected and removed. After removing the face, both the signer’s hands are detected by using the skin color-based segmentation technique. After that, a feature set composed of one temporal and three spatial features is derived. This feature set is used to train the HMM which is finally used for separating out the “sign” and “*me*” frames.

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TABLE 2. Continuous Sign Language Spotting Results

Videos	Sign sequence	Number of sign segments	Ground-truth for signs	Spotted sign segments
1	Adopt-Advice	2	Adopt (37-100) Advice (188-248)	Adopt (40-107) Advice (175-252)
2	Camping-Can	2	Camping (10-88) Can (138-162)	Camping (4-91) Can (139-162)
3	Building-Busy	2	Building (28-85) Busy (170-238)	Building (33-90) Busy (168-258)
4	All-Accident	2	All (10-80) Accident (146-171)	All (11-97) Accident (140-170)

TABLE 3. Spotting Rate of Continuous Sign Sequences

Videos	Sign sequence	Number of sign segments	Average R (%)	Overall R (%)	Overall R (%) (without PCA)
1	Adopt-Advice	2	82	83	78
2	Camping-Can	2	89		
3	Building-Busy	2	84		
4	All-Accident	2	78		

observe the state of each frame of a sign sequence video, the length of the Viterbi path is equal to the number of frames in the video.

Quantitative evaluation of the proposed method is carried out by correct spotting rate [4] which is calculated as

$$R = \frac{C}{N} \times 100 \quad (3)$$

where C is the number of correct spottings and N is the number of test signs. The correct spotting c_s , is defined as [4]

$$c_s = \begin{cases} \text{True, if } (\Delta S + \Delta E) < 10 \\ \text{False, otherwise} \end{cases} \quad (4)$$

where $\Delta S = |g_s - s_s|$ and $\Delta E = |g_e - s_e|$. s_s and s_e are the start and end frame numbers of the spotted sign, respectively, and g_s and g_e are the start and end frame numbers of the ground truth of the spotted sign, respectively.

Table 2 shows the sign spotting results for four sign-sequence videos articulated by Signer 1 and Signer 2. From the table, it is observed that for sign-sequence "Camping-Can," both the signs are spotted correctly. But, in the case of sign-sequence "Adopt-Advice," sign "Adopt" is correctly spotted, but the sign "Advice" is not spotted correctly. Altogether, we have considered six different sign-sequences consisting of two signs in each sequence articulated by eight signers. The average spotting rates for four such sign sequences are shown in Table 3. The overall spotting rate is found to be about 83%, which is better than the spotting rate without using PCA (78%). The spotting rate is superior to the proposed method based on support vector machine by Yang and Lee [3]. In [3], they proposed a sign and fingerspelling spotting method using hierarchical steps. First, a two-layer conditional random field (CRF) is applied to discriminate candidate segments of signs and fingerspellings. After detecting the candidate segments, their hand shapes are verified using BoostMap embeddings (BMEs) and support vector machine (SVM). Finally, the motions of fingerspellings were verified with the CRF. They used both hand-motion and hand-location-based features. They achieved 78% and 83% spotting rates for signs using SVM and BME, respectively. Also, Jian *et al.* [23] proposed a transformer-based network called sign-lookup for continuous SL spotting. Although their work achieved a spotting rate of 96%, their system requires two types of input videos viz., isolated sign videos (query) and continuous sign videos (target). But our system does not require such query signs and can spot the signs from the continuous sign sequence videos irrespective of the signs present.

IV. CONCLUSION

In this letter, we use HMM in a novel way for continuous SL spotting. A two-state HMM with Gaussian emission probability is trained with the features extracted from the sign sequence video, followed by the decoding of the state-sequence for spotting the sign components from the sign sequence. The fusion of features obtained from both the pixel as well as compressed domain analysis is done to exploit the benefits of individual domains. From the experimental results, it is seen that

the proposed method can spot the sign frames with an average spotting rate of about 83%.

The work can be further extended for the spotting of sentences containing more than two signs. Once the spotting is done, HMM models for each sign can be used for the recognition of the signs. The work has lots of scopes for further investigation.

REFERENCES

- [1] W. Gao, G. Fang, D. Zhao, and Y. Chen, "A Chinese sign language recognition system based on SOFM/SRN/HMM," *Pattern Recognit.*, vol. 37, no. 12, pp. 2389–2402, 2004.
- [2] R. Yang, S. Sarkar, and B. Loeding, "Handling movement epenthesis and hand segmentation ambiguities in continuous sign language recognition using nested dynamic programming," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 3, pp. 462–477, Mar. 2010.
- [3] H.-D. Yang and S.-W. Lee, "Simultaneous spotting of signs and fingerspellings based on hierarchical conditional random fields and boostmap embeddings," *Pattern Recognit.*, vol. 43, no. 8, pp. 2858–2870, 2010.
- [4] H.-D. Yang, S. Sclaroff, and S.-W. Lee, "Sign language spotting with a threshold model based on conditional random fields," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 7, pp. 1264–1277, Jul. 2009.
- [5] S. G. Azar and H. Seyedarabi, "Trajectory-based recognition of dynamic persian sign language using hidden Markov model," *Comput. Speech Lang.*, vol. 61, pp. 1–10, 2020.
- [6] M. Mohandes, M. Deriche, U. Johar, and S. Ilyas, "A signer-independent arabic sign language recognition system using face detection, geometric features, and a hidden Markov model," *Comput. Elect. Eng.*, vol. 38, no. 2, pp. 422–433, 2012.
- [7] P. Kumar, H. Gauba, P. P. Roy, and D. P. Dogra, "Coupled HMM-based multi-sensor data fusion for sign language recognition," *Pattern Recognit. Lett.*, vol. 86, pp. 1–8, 2017.
- [8] W. Yang, J. Tao, and Z. Ye, "Continuous sign language recognition using level building based on fast hidden Markov model," *Pattern Recognit. Lett.*, vol. 78, pp. 28–35, 2016.
- [9] R. Elakkiya and K. Selvamani, "Enhanced dynamic programming approach for subunit modelling to handle segmentation and recognition ambiguities in sign language," *J. Parallel Distrib. Comput.*, vol. 117, pp. 246–255, 2018.
- [10] J. Tang, H. Cheng, Y. Zhao, and H. Guo, "Structured dynamic time warping for continuous hand trajectory gesture recognition," *Pattern Recognit.*, vol. 80, pp. 21–31, 2018.
- [11] K. L. Cheng, Z. Yang, Q. Chen, and Y.-W. Tai, "Fully convolutional networks for continuous sign language recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 697–714.
- [12] R. Elakkiya, P. Vijayakumar, and N. Kumar, "An optimized generative adversarial network based continuous sign language classification," *Expert Syst. Appl.*, vol. 182, 2021, Art. no. 115276.
- [13] R. Dugad and U. B. Desai, "A tutorial on hidden Markov models," *Signal Process. Artif. Neural Netw. Lab., Dept. Elect. Eng., Indian Inst. Technol., Bombay, India, Tech. Rep.*, 1996.
- [14] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proc. IEEE*, vol. 77, no. 2, pp. 257–286, Feb. 1989.
- [15] P. Viola and M. Jones, "Rapid object detection using a boosted cascade of simple features," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 1, 2011, pp. 1–9.
- [16] D. Kelly, J. McDonald, and C. Markham, "Recognizing spatiotemporal gestures and movement epenthesis in sign language," in *Proc. IEEE 13th Int. Mach. Vis. Image Process. Conf.*, 2009, pp. 145–150.
- [17] G. A. Rao and P. Kishore, "Selfie video based continuous indian sign language recognition system," *Ain Shams Eng. J.*, vol. 9, no. 4, pp. 1929–1939, 2018.
- [18] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 1, 2005, pp. 886–893.
- [19] A. K. Talukdar and M. K. Bhuyan, "Movement epenthesis detection in continuous fingerspelling from a coarsely sampled motion vector field in H.264/AVC video," in *Proc. IEEE Recent Adv. Intell. Comput. Syst.*, 2018, pp. 26–30.
- [20] S. M. Pereira and G. Leslie, "Hypothesis testing," *Aust. Crit. Care*, vol. 22, no. 4, pp. 187–191, 2009.
- [21] C. O. S. Sorzano, J. Vargas, and A. P. Montano, "A survey of dimensionality reduction techniques," 2014, *arXiv:1403.2877*.
- [22] V. Athitsos *et al.*, "The american sign language lexicon video dataset," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. Workshops*, 2008, pp. 1–8.
- [23] T. Jiang, N. C. Camgöz, and R. Bowden, "Looking for the signs: Identifying isolated sign instances in continuous video footage," in *Proc. 16th IEEE Int. Conf. Autom. Face Gesture Recognit.*, 2021, pp. 1–8.